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LXVI. *The Remanent Magnetism of Some Sedimentary Rocks in Britain*

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Abstract

This paper describes a survey which was carried out to measure the natural remanent magnetism of Triassic sediments in England. Rocks from nine separate sites distributed over a wide area were found to have consistent polarizations, being magnetized in an approximately North East–South Westerly direction with dips significantly less than that of the present earth's field. Similar results were obtained for sediments of the Carboniferous and Old Red Sandstone periods taken from two other sites. Approximately half of the specimens showed reverse polarization. Tests were also made to measure the magnetic stability of the rocks. The possible implications of the results are discussed.

§ 1. Introduction

The remanent magnetism of sedimentary rocks has been studied during the past two decades and has yielded important information concerning the prehistory of the earth's magnetic field.† The most reliable data obtained in this way to date have come from the investigation of the natural magnetization of varved clays. Johnson, Murphy, and Torreson (1948) in particular made a very complete set of measurements on a continuous vertical sequence of New England clays, extending back in age to 15 000 B.C., and found that the directions of polarization show regular variations about geographic North ; while Nagata (1951), Kawai (1951) and others observed similar directions of magnetization in certain Japanese clays dating back to the early Miocene period.

The information concerning earlier sediments is more scanty, but the data published to date indicate that the majority have magnetic polarizations lying reasonably close to the axial dipole field, and that while a number show oblique magnetization, the distribution is not such as to indicate a significant deviation towards any other preferential direction.

In October, 1952, we undertook a study of the remanent magnetism of certain sediments of the Triassic system. The first set of samples from North West Cheshire showed a consistent direction of magnetization which differed appreciably from that of the present earth's field. A more comprehensive investigation extending first to other parts of

* Communicated by Professor P. M. S. Blackett, F.R.S.

† For a general review of magnetic measurements made on igneous and sedimentary rocks, see Nagata (1953).

Cheshire, and later over a wider area, showed that the majority of English rocks of the same geological system had similar polarizations. This survey has since been extended to rocks of earlier ages, and the detailed results are set out below.

§ 2. Method of Collection and Measurement

The rocks were taken from exposures in road cuttings, quarries, the beds and banks of rivers, and sea cliffs. A photograph of a typical exposure is shown in Plate 18. Each sample* consisted of a slab a few inches in thickness, and was oriented *in situ* before removal. Whenever possible at least two samples were taken from positions a few feet apart at each site. These slabs were cored in the laboratory with a diamond trepanning tool, and the cylindrical cores were cut with a diamond wheel into disc-shaped specimens $1\frac{3}{8}$ in. diameter and $\frac{3}{16}$ in. in thickness. The plane faces of the discs were cut parallel to the original bedding surfaces of the rock and the upper surface of each was marked with an arrow drawn perpendicular to the strike, to indicate the downward direction of tilt† of the strata.

The direction and intensity of magnetization of the specimens were measured by means of a magnetometer of the type designed initially by Blackett (1952). This instrument consists essentially of a pair of carefully astaticized magnets suspended on a torsion fibre, the earth's magnetic field at the magnet system being annulled by three pairs of mutually perpendicular Helmholtz coils. Its sensitivity was such that a vertical field gradient of 10^{-7} oersteds per cm produced a scale deflection of 1 mm. The specimen was mounted with its axis vertical on a horizontal turntable which could be raised and lowered beneath the magnetometer. The method of using this magnetometer to measure the horizontal direction and intensity of magnetization by bringing the turntable up to the magnet system in four successive positions, and the 'off centre' method of measuring the dip have been described in detail elsewhere (Blackett 1952).

The rocks had intensities of magnetization ranging from 6×10^{-7} to 150×10^{-7} c.g.s. units. The standard error of a single determination of declination was 3°, and of dip 4°.

All directional measurements were made relative to the tilt arrow marked on the upper face of the specimen. No attempt was made to correct for the tilt of the strata, and the declinations and dips quoted below are referred to the bedding planes, the zero of declination being the intersection of the bedding surface of the parent rock with the plane of the earth's magnetic meridian.

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* Throughout this paper the slabs initially taken from the parent rock will be referred to as 'samples' and the term 'specimen' will be used for the individual cylindrical discs, a number of which were cut from each sample.

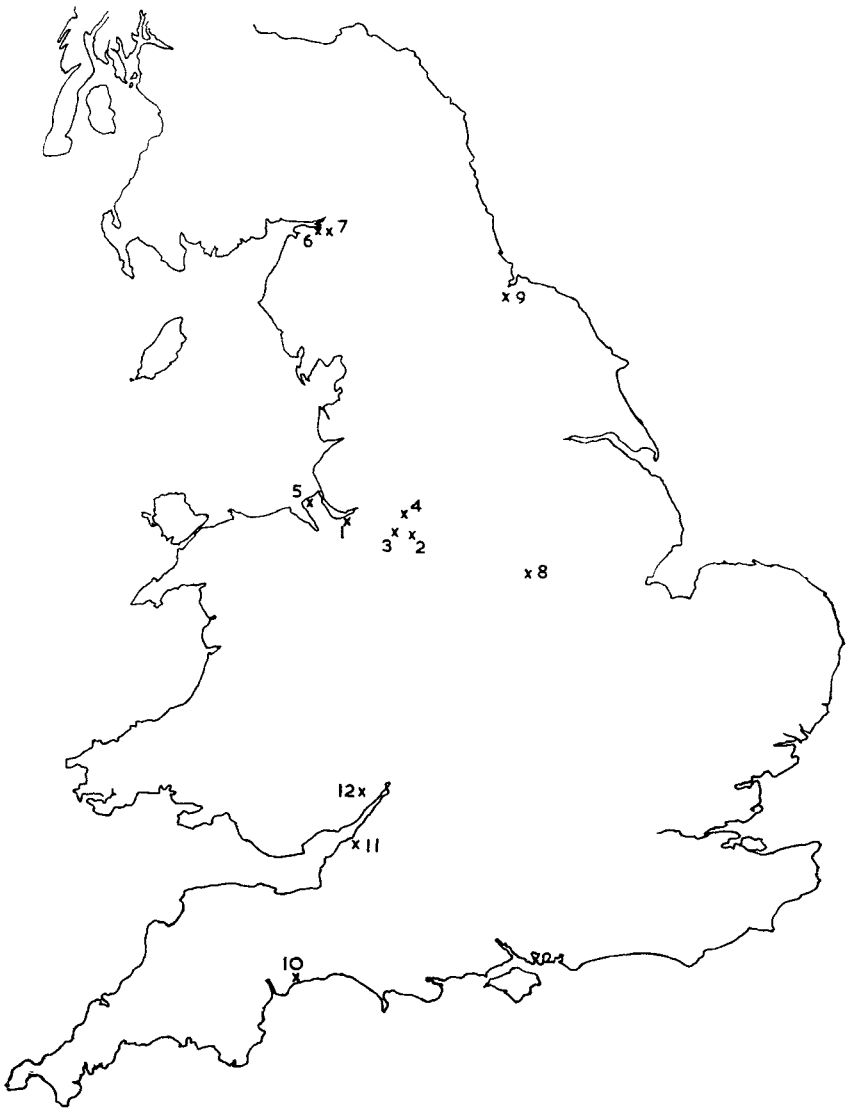
† See footnote at bottom of table 1.

§ 3. Results

(i) *The Remanent Magnetism of Triassic Sediments in England*

We undertook these investigations on the supposition that the remanent magnetic moment of certain sedimentary rocks was acquired during

Fig. 1



Map showing positions of sites from which specimens were taken.

deposition by alignment of the ferromagnetic particles along the earth's magnetic field, the direction of polarization having been preserved unchanged throughout subsequent geological ages up to the present time.

This implies that the sediments were laid down under tranquil conditions and did not subsequently undergo undue deformation, and that the original magnetic constituents were sufficiently stable to resist any demagnetizing effects to which they may have been exposed since deposition. Among the Triassic rocks which we examined, and for which it appeared likely that these conditions might have been fulfilled, were certain fine red sandstones occurring at Frodsham, Cheshire (site (1), fig. 1). These deposits, which had well defined uniform bedding planes with a tilt of only 5°, were exposed in a newly made road-cutting, and samples were taken from the same horizon at 20 yard intervals over a horizontal distance of 100 yards. Measurements made on a number of specimens cut from each sample showed a consistent direction of magnetization, approximately 40° East of the present magnetic North, with a dip somewhat lower than that of the present earth's field. A second set of samples taken from other horizons at the same site, over a vertical range of 12 ft., yielded similar results, and the mean direction for all specimens was found to be 41° East of magnetic North in declination, and 52° downward in inclination.

These sandstones occur close to the base of the Keuper Marl series, and have an age of the order of 180 million years. We decided to examine other red sandstones of the same period taken from sites several miles apart in Cheshire, at Congleton (2), Holmes Chapel (3), and Wilmslow (4). The measured directions and intensities of magnetization are shown in table 1 and the directions of polarization are plotted on the projection in fig. 2. The results for site (3) correspond closely with those for site (1), but the polarization of the rocks of site (4) was in almost exactly the reverse direction. The samples from site (2), on the other hand, were reversed in declination but had almost zero dip. It appears, therefore, that the Triassic rocks throughout Cheshire are magnetized in an approximately North–East to South–West line, the dips being somewhat variable but on the whole considerably less than that of the present earth's field, and that in some cases the direction of magnetization is reversed. This consistency in the direction of the axis of magnetization appears more striking when we consider the probable difference between the ages of the rocks from the four sites. Thus, Wilmslow and Holmes Chapel are considerably nearer to the centre of the Cheshire Basin than the other two sites, and the horizons from which the samples were taken may well differ by 1500 feet.

In view of this uniformity of magnetization, it appeared expedient to extend our investigations to other parts of the Trias outside the Cheshire Basin, and we examined samples of similar red sandstones of the same geological series from the Wirral Peninsula (5), Beaumont, Cumberland (6), Carlisle (7), Nottingham (8), the River Leven at Rudby, North Yorkshire (9), and Sidmouth, Devon (10). The results shown in table 1 and fig. 2 for sites (5) to (9) are in general agreement with those obtained for the previous locations. The rocks from Sidmouth (site (10)), however,

Table 1. Results Obtained for Rocks of Keuper Marl Series

Site no.	Locality	Nat. grid reference	Mean direction of downward tilt*	Mean angle of tilt	Total no. of samples taken	Total no. of specimens measured	Intensity of magnetization $\times 10^{-7}$ c.g.s.	Mean declination E. of M.N.†	Mean dip*	Radius of 50% circle of confidence‡
1	Frodsham, Cheshire	352·7 E ; 378·2 N	69½°	5½°	12	191	7·2	41°	+52° down	2·5°
2	River Dane, Cheshire	388·7 E ; 368·9 N	26°	8½°	6	33	9·2	236°	+9° down	1·5°
3	Holmes Chapel, Cheshire	377·8 E ; 367·6 N	186½°	7°	2	38	12·0	45°	+39° down	1·5°
4	River Bollin, Cheshire	381·3 E ; 382·8 N	260°	4°	2	37	35·2	208°	−28° up	1·5°
5	Wirral Peninsula	325·3 E ; 387·3 N	318½°	4½°	6	39	34·4	34°	+27° down	2·0°
6	Beaumont, Cumberland	335·1 E ; 560·0 N	172½°	36½°	2	37	12·8	246°	−27° up	1·5°
7	Carlisle, Cumberland	337·8 E ; 556·6 N	236°	5°	3	37	11·2	44°	+28° down	3·0°
8	River Trent, Notts.	464·8 E ; 340·0 N	84°	4°	8	119	6·3	225°	−16° up	1·5°
9	River Leven, N. Yorks.	446·1 E ; 507·1 N	107°	15°	2	9	140·5	26°	+22° down	1·5°

Table 2. Results Obtained for Rocks of Old Red Sandstone and Upper Coal Measure Series

Site no.	Locality	Nat. grid reference	Mean direction of downward tilt*	Mean angle of tilt	Total no. of samples taken	Total no. of specimens measured	Intensity of magnetization $\times 10^{-7}$ c.g.s.	Mean declination E. of M.N.†	Mean dip*	Radius of 50% circle of confidence‡
11	Frampton Cotterill, Gloucester	366·8 E ; 183·2 N	73°	19°	1	14	4·7	44°	+35° down	5·0°
12	Mitcheldean, Gloucester	367·2 E ; 218·5 N	277°	42°	3	39	6·7	244°	−22° up	5·5°

* Throughout this paper the term ' tilt ' refers to the inclination of the bedding planes. The term ' dip ' is used to indicate the inclination of the remanent magnetization and is measured relative to the plane of the strata.

† All azimuthal directions are measured in degrees East of Magnetic North, in the plane of the strata.

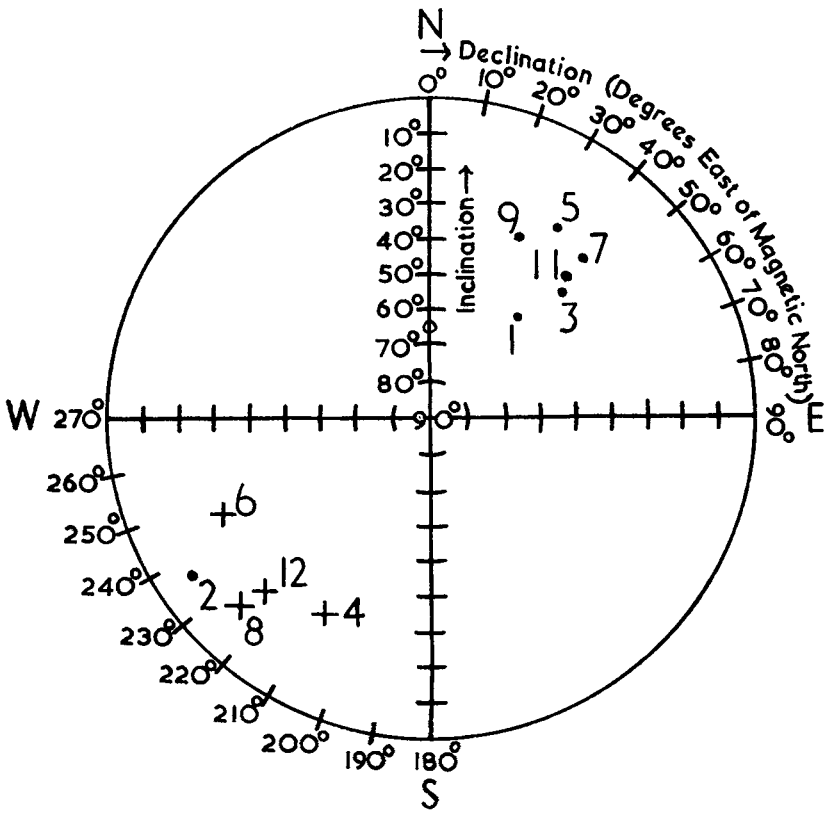
‡ This has been computed by the method devised by Fisher (1953).

showed random polarizations, and were found to be unstable in the earth's field. The possible implication of this instability will be discussed in § 5.

(ii) *The Remanent Magnetism of Carboniferous and Devonian Rocks*

Following the measurements on Triassic rocks we embarked on a preliminary survey of the remanent magnetism of sediments of other geological periods. This is still continuing, but measurements have been made to date on two sets of samples. The first of these were of Pennant

Fig. 2



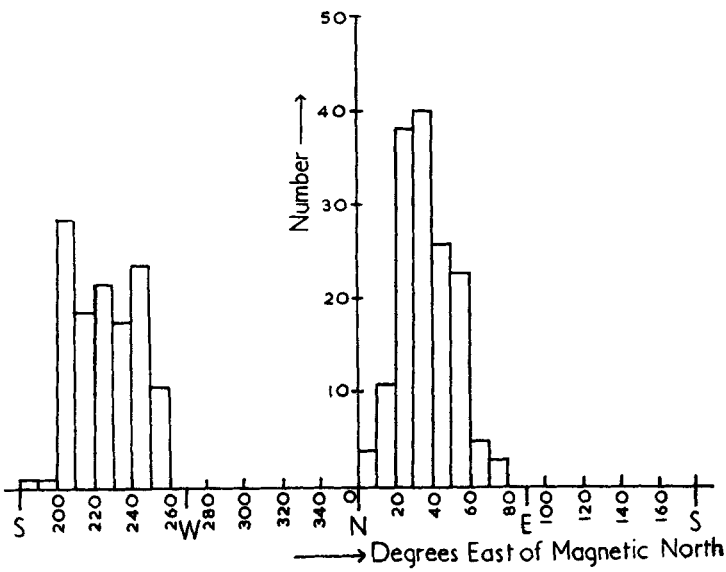
Polar projection showing mean directions of polorization of specimens from various sites.

● Downward dip
+ Upward dip
○ Present magnetic North

Sandstone of the Upper Coal Measures occurring at Frampton Cotterill, Gloucestershire (site (11)). The second from Mitcheldean in Gloucestershire (site (12)), were of Lower Old Red Sandstone (Brownstone series) approximately 300 million years old. The results, which are included in fig. 2 and table 2, indicate that these rocks have the same general axis of magnetization, the first showing a North Westerly declination with a downward dip and the second being polarized in the reverse direction.

The most notable feature of the foregoing results is the consistency in the declination measurements. This becomes particularly evident when a combined histogram is drawn, as in fig. 3(*a*),* showing the frequency distribution of the declinations of the individual specimens from all the sites listed in tables 1 and 2. It can be seen that they fall into two approximately numerically equal groups, corresponding to azimuthal directions of magnetization of 37° and 226° East of present magnetic North. The inclinations are proportionately more scattered, but the distribution in fig. 3(*b*) shows two well defined maxima at +33° and −27°. The fact that the mean upward dip was greater than the downward is mainly due to the specimens from sites (2) and (8), where the inclinations were scattered and were distributed about zero.

Fig. 3(*a*)



Frequency distribution of declinations.

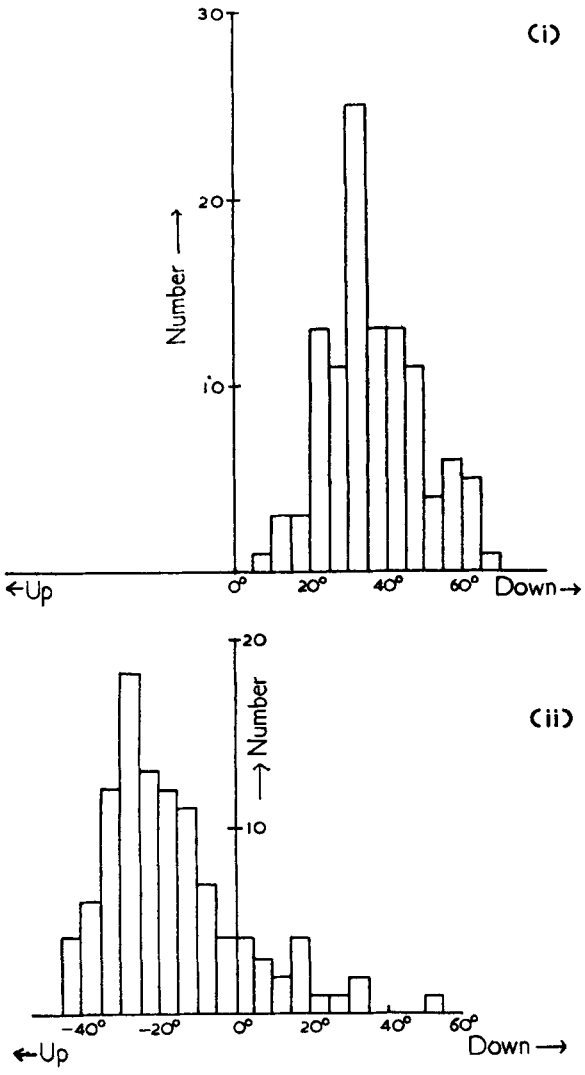
§ 4. Stability of Magnetization

The fact that all the rocks with the exception of those from site (10) show such marked uniformity in direction of magnetization, and that this direction differs significantly from that of the present earth's field, evidently indicates some degree of magnetic stability over a considerable period of time. Particularly good evidence of this is afforded by the measurements made on samples from Beaumont, Cumberland (site (6)).

* Since different numbers of specimens were taken from the different sites, the following procedure was adopted to obtain the histograms shown in fig. 3. Separate distributions were first found for each of the 11 individual sites. These were normalized to correspond to a total of 25 specimens from each site, and were then added.

These rocks, which were found in the bed of the River Eden, were tilted downwards towards the South at an angle of 35°, the strike being almost perpendicular to the magnetic meridian. The figures in table 1 show that

Fig. 3 (*b*)



Frequency distribution of dips.

(i) Dips of specimens showing N.E. declinations.

(ii) Dips of specimens showing S.W. declinations.

the direction of magnetization, referred to the bedding planes, corresponds reasonably well with the directions found for other reversed specimens taken from sites where the bedding was nearly horizontal. This suggests

that the magnetization was acquired before tilting, and that the direction of polarization, which is almost perpendicular to that of the present earth's field, has since been preserved. It is not possible to obtain an accurate estimate of the time which has elapsed since the tilting occurred, nor has the mechanism by which it was effected been ascertained, but the extent of the erosion required to reduce the beds to their present topography indicates that it must have taken place at least some thousands of years ago. It therefore appears highly probable that the remanent magnetization of at least some of these Triassic Sandstones can remain substantially unchanged in the earth's field over periods of this order. That it remains stable over short periods was confirmed in the laboratory, where specimens were stored with their magnetic axes oriented in random directions. With the single exception of the rocks from site (10), measurements made on numerous specimens at intervals of several weeks gave values for the directions and intensities of magnetization which agreed within the probable error of measurement. However, all specimens from site (10) were markedly unstable in the laboratory, and the direction of magnetization could be substantially changed by changing the orientation relative to the earth's field for a period of a few hours. Measurements on these samples have therefore not been included in the results quoted in this paper.

In view of the fact that the rocks may have been subjected in the past to fields somewhat greater than the present earth's field, and may, furthermore, have been heated by burial, we carried out a series of magnetic stability tests on specimens from all sites excluding site (10) in the laboratory. A brief résumé of our results is given below.

(i) *The Effect of Steady Magnetic Fields at Room Temperature*

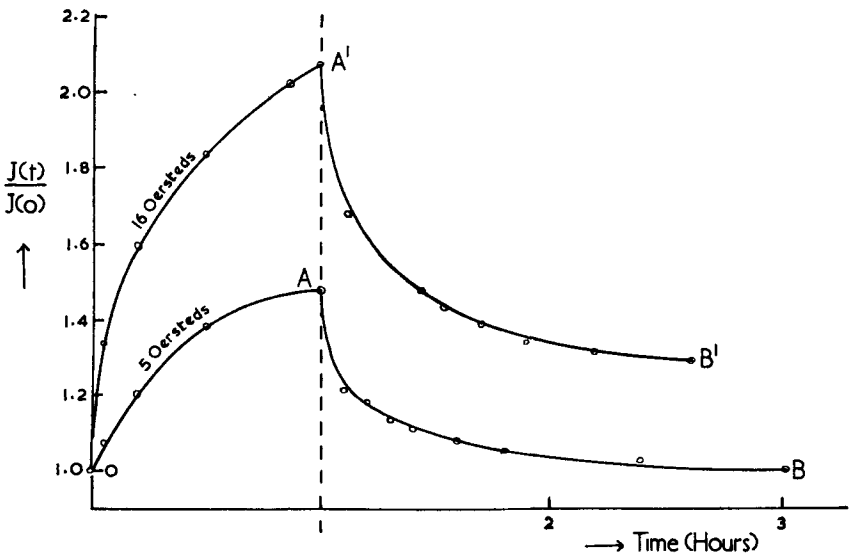
Although the magnetization of the specimens remained stable in the earth's magnetic field, it could readily be changed by applying steady fields of the order of a few oersteds. A notable feature of the changes produced by d.c. fields was that they were found to depend not only on the field strength, but also on the length of time during which it was applied. They were moreover reversible, and on removing the applied field, provided this was not too big, the magnetic moment of the specimen returned slowly towards its initial value. A typical example of this time dependence is depicted in fig. 4. A particular specimen was first placed in a field of 5 oersteds for one hour, with its axis of remanent magnetism directed along the field. Measurements made at intervals of several minutes showed that the intensity of magnetization $J(t)$ increased during the whole of this period, and the curve OA shows the ratio of $J(t)$ to the initial intensity $J(o)$ plotted as a function of time. After one hour the magnetizing field was removed, and during the following two hours the ratio $J(t)/J(o)$ reverted to unity, in the way shown by the curve AB. The curve OA′B′ shows the results obtained for the same specimen when a magnetizing field of 16 oersteds was applied for the

same length of time. In this case $J(t)/J(o)$ did not return to unity after the removal of the field but to a final value $J(\infty)/J(o)=1{\cdot}26$, which subsequently remained substantially constant over a period of months.

Tests similar to these were made to find the effects of steady magnetic fields on numerous samples, and the results can be summarized briefly as follows :

(*a*) The magnetization remains unchanged provided that the applied field strength is less than a certain critical value H_c. This critical value differs from sample to sample but never exceeds 5 oersteds and in some cases is as low as 0·5 oersteds. The instability of the samples from site (10) appears to be due to the fact that the critical value is less than the earth's field in this case.

Fig. 4



Isothermal magnetization and decay in steady fields.

(*b*) When the applied field H is greater than H_c, the intensity of magnetization changes rapidly at first, and then more slowly, finally attaining a constant value after a period of the order of a few hours. When the field H is removed, the induced magnetism decays in a similar period. For small values of $(H-H_c)$ the intensity $J(t)$ finally reverts to the initial value $J(o)$, but for sufficiently large values of $(H-H_c)$ it approaches a constant value $J(\infty)$ not equal to $J(o)$. In the latter case, therefore, there is a permanent induced magnetization $J(H)=J(\infty)-J(o)$ which is a function of the applied field strength H.

(ii) *The Effect of Alternating Magnetic Fields*

In contrast to their magnetic instability when exposed to d.c. fields, the rocks show a high degree of stability when exposed to alternating fields. In a series of a.c. demagnetizing tests made on a number of specimens taken from different sites, using a frequency of 50 c/s, the

magnetic moment was found to remain unchanged even when the strength of the demagnetizing field exceeded 300 oersteds.

Moreover, when a specimen had acquired a permanent induced magnetization $J(H)$ by the application of a sufficiently large d.c. field in the way described in (i) above, this could be removed, and the rock reduced to its original magnetic condition, by the application of relatively low a.c. fields.

(iii) *Temperature Effects*

It was found that the effects of steady magnetic fields described in (i) were sensitive to relatively small changes in the ambient temperature, the critical field strength H_c required to change the magnetization being reduced in many cases by a half when the temperature was increased from 16°C to 45°C. At temperatures greater than 80°C H_c invariably became so small that the magnetization was unstable in the earth's magnetic field. The rate of rise and decay of the induced magnetism could also be accelerated, and the amplitude of the decay curve could be increased, by increasing the temperature.

The ease with which the magnetization of these rocks can be changed by small d.c. fields and subsequently reduced again to its original value, combined with the high stability of the initial remanent magnetism in a.c. fields, can best be explained on the assumption that they contain at least two magnetic components with very different properties. We suppose that the first or soft component accounts for the effects described in (i) and (iii) above. It is not affected by the earth's field at room temperature and is normally unmagnetized in the native rock (the exception to this being the rocks from site (10), for which the critical field H_c required to change the magnetization is evidently less than that of the earth). Under normal conditions it is unstable in steady fields of the order of a few oersteds, and the stability can be decreased by increasing the temperature. It can, moreover, be demagnetized easily in a.c. fields.

The second, or hard, component, which is less easily changed, is responsible for the remanent magnetism found in the native rocks.

§ 5. Laboratory Deposition Tests

It has already been pointed out that the first Frodsham sandstones found at site (1) had all the characteristic features of quiet sedimentation. In order to obtain some indication of the magnetization that they might have acquired on deposition, a number of rocks from this site were powdered and redeposited in the laboratory. Although we do not propose to give a complete description of these experiments in the present paper, certain unpublished results due to Mr. J. L. Lloyd are germane to the discussion in the following section and may be quoted.

The powdered rock was redeposited in still water in the earth's magnetic field. The water was then decanted and the sediment was allowed to dry. Specimens taken from this deposit were found to have a magnetic polarization which coincided in declination with the earth's field, but which had a dip some 8° lower than the magnetic inclination. The

intensity of magnetization was about three times as great as that of the original rock before powdering. On the basis of these results, the assumption that the rocks took up their magnetization on deposition appears feasible. The lower magnetic intensity of the natural specimens could be accounted for by a rise in the strength of the earth's field since Triassic times, or by turbulence during deposition. Alternatively it may be due to the decay of the remanent magnetism with time, and in this connection it may be worth noting that Königsberger (1932) and Nagata (1943) have found decay times comparable to this in the remanent magnetism of igneous rocks.

Many of the Triassic rocks collected later did not show such regular banding as those originally found at Frodsham. Certain samples had obviously been deposited under excessively turbulent conditions, while others showed sun cracks and breccias. That the magnetic polarization of rocks such as these should have been taken up on deposition appears improbable. In this connection, however, certain other laboratory tests are of interest. A quantity of the powdered rock was deposited in water in zero magnetic field, and the water was then decanted. The sludge remaining in the bottom of the tank still contained about 75% by weight of water, which dried out in the course of a few days. By applying a magnetic field at different stages of the drying process it was found that a magnetization of comparable intensity to that of the parent rock could be induced so long as the water content of the deposit was greater than 50% by weight.

§ 6. Discussion of Results

The measurements described in § 3 appear to show conclusively that Triassic sediments throughout England have a preferential direction of remanent magnetization, differing significantly from that of the present earth's field, and that this direction is shared by at least some of the Carboniferous and Devonian sediments. They also show that the magnetic polarization is reversed in approximately 50% of the rocks examined. This consistency evidently has some physical significance, and while it is not possible, on the basis of such a limited survey, to draw any definite conclusions as to its cause, sufficient data is now available to justify a tentative consideration of the more obvious explanations.

Since our interpretation of the results will be influenced by our view as to the way in which the rocks acquired their magnetization, this question will first be considered in somewhat greater detail. The uniformity in the direction of the magnetic axis and the problem of the magnetic reversals will then be discussed separately.

(i) *The Origin of the Remanent Magnetism*

(*a*) *Magnetization Before Cementation*

The information gleaned from the deposition tests described in § 5 indicates that the rocks may well have become magnetized before cementation. This could have occurred in some cases on deposition, or,

in view of the magnetic mobility of the saturated sediments, at some time later, even perhaps during the resaturation of the unconsolidated deposits.

(*b*) *Magnetization by Heating*

In considering the origin of the remanent magnetism of these rocks, it is impossible, on the basis of the data at present available, to rule out the alternative possibility of its having been acquired by heating due to burial deposition. The total thickness of the sediments later than the Trias in England may amount to some 10 000 feet (O. T. Jones, private communication). Taking this as the maximum depth to which the New Red Sandstones could have been buried, we obtain an upper temperature limit of the order of 100°C. While the soft magnetic component present in the rocks would be magnetized in the earth's field at this temperature, it seems unlikely, in view of the laboratory tests made so far that the hard component would be affected. However, conclusive evidence on this point can only be obtained by a more extensive study of the composition and physical properties of the magnetic minerals present.

(ii) *The Direction of the Magnetic Axis*

In the following discussion we will assume that the mean direction of the earth's dipole axis has corresponded in the past with the axis of rotation. Modern theories of the earth's field (Bullard 1949, Runcorn 1954) support this view, and as we have already pointed out, there is some weight of experimental evidence which suggests that the general direction of the earth's field has coincided reasonably closely with that of an axial dipole during the past 17 000 years (Johnson, Murphy and Torreson 1948) and probably for the past 20 million years (Nagata 1951, Kawai 1951).

The mean direction of polarization of the rocks examined differs by 34° from geographical North. It appears unlikely that a deviation as large as this and of such wide-spread consistency, could be caused by a difference between the local magnetic declination and true North at the time of magnetization. It is true that the apparent North Easterly declination may have been enhanced by other causes. Thus if the earth's field had a sufficiently steep dip at the time when the rocks were magnetized, a small departure of its direction from the plane of the geographic meridian could give rise to a large Easterly declination. A similar result might be produced if the rocks had been tilted about an East–West axis since the time when they took up their magnetization. It might be argued that the present North–East to South–Westerly direction of the remanent magnetism could be due to either of these effects, or to the two working in conjunction, but a numerical consideration of the possible combinations of magnetic inclination and the tilt of the strata shows this to be improbable. The geological evidence indicates that the tilting of the rocks could not have amounted to more than a few degrees, and the dip of the earth's field required to change the apparent declination by an appreciable factor would in consequence be considerably

greater than that found experimentally in the rocks. A still more cogent objection to any explanation based on the foregoing arguments is presented by consideration of the relative time scales involved. The time period of the secular changes in the earth's field are of the order of hundreds of years, and it appears inconceivable, however the remanent magnetism was acquired, that rocks distributed over such a wide area and having ages ranging from the Devonian to the Triassic should all have acquired their polarization at times when the field had the same general direction.

It is possible to envisage ways in which the direction of the magnetization of the rocks might have been influenced by mechanical forces acting during deposition. For example, if the deposits were laid down as Aeolian sands and if the magnetic particles were needle or disk shaped, they might be expected to orient themselves in some preferred direction relative to the prevailing wind. A similar result could be produced by current bedding in water. It seems impossible, however, that the magnetism of the rocks from all eleven sites could have been influenced by effects such as these. The probability of uniform current conditions of any kind having prevailed over the whole area for the required length of time is evidently negligible.

Another conceivable way in which the preferential direction of magnetization may have been acquired is by the action of shearing strains on the rock after consolidation. Such distortion might be expected to lead to the orientation of the magnetic mineral grains in a preferential direction. There is, however, no geological evidence of such a distortion having occurred in any of the rocks examined.

Finally, it seems therefore that the most likely explanation of the observed horizontal direction of magnetization of the sediments studied is that the whole of the land mass which now constitutes England has rotated clockwise through 34° relative to the earth's geographical axis. This movement must have occurred since the rocks acquired their magnetization, which was probably at or soon after the time of deposition (150 to 200 million years in the case of the Triassic sediments), but conceivably much later.

If such a rotation of England has occurred, it could have been a local movement of part only of the earth's crust, or alternatively the earth's mantle could have moved as a rigid whole relative to the geographical poles. The first hypothesis would consider the rotation either as a purely local movement or as part of a drift of large continental land masses. The second would adduce pole wandering as the operative mechanism. By measuring the direction of the remanent magnetism of rocks of different geological ages in different parts of the world, it should be possible to distinguish between these hypotheses.

It has been shown that the rocks also have a mean magnetic dip markedly less than that of the axial dipole field in the latitude of England (68°). Some part of this difference may be accounted for by deformation

or compaction subsequent to deposition, but when due allowance has been made for these effects it still seems probable that the magnetic dip at the time of magnetization was appreciably less than today. If this is true, it would imply an increase in magnetic latitude, since magnetization and consequently, on the assumption of an average coincidence between the magnetic and rotational axes, an increase in geographical latitude.

(iii) *Magnetic Reversals*

Of the rocks from the eleven sites studied, those from six are magnetized in a roughly North-Easterly direction with a mean downward dip and those from the other five in a South-Westerly direction with a mean upward dip. As seen from tables 1 and 2 and fig. 3, the angle between the average magnetic polarizations of the two sets is quite close to 180°, and no specimens are magnetized in directions nearly perpendicular to this. We have been unable to find any significant difference between the magnetic and other physical properties of the directly and reversely polarized specimens.

Although this appears to be the first time that this effect has been demonstrated so definitely in sedimentary rocks, an exactly similar phenomenon has been observed with igneous rocks by Bruckshaw and Robertson (1949), Roche (1950), Hospers (1951, 1953). In particular, Bruckshaw and Robertson measured the remanent magnetism of a number of tholeiite dykes of the Tertiary Epoch occurring in Northern Britain, and found many examples of reverse remanent magnetization, while Hospers in a series of measurements on recent Icelandic lava flows found approximately equal numbers of sites showing direct and reverse polarizations with very few intermediate values. In none of the rocks examined was it possible to detect any physical or chemical difference between the direct and reversed specimens. In each of these cases the samples were taken from a limited area, and the reversals might conceivably have been caused by a purely local reversal of the earth's field at the time of magnetization. However, since it is very difficult to envisage a mechanism which would change the direction of the earth's field locally through almost exactly 180° without showing a series of intermediate directions, all three authors concluded that if a field reversal did in fact occur it must have been world wide, and have amounted to a rather sudden reversal of the main dipole field.

On the other hand Graham (1953) holds that the hypothesis of repeated reversals of the mainfield is not yet proved, and that some physical or chemical effect of the type envisaged in Néel's four alternative hypotheses (Néel 1949, 1952) may have been the main cause of the adverse magnetization of rocks. Such self reversals have been observed in igneous rocks by Nagata (1951) although the phenomenon appears to be rare and his specimens cannot be considered at all typical.

Returning to our own results it seems evident that if the present magnetization of these Palaeozoic and Mesozoic rocks was acquired by orientation of the magnetic particles in the unconsolidated sediments, the results

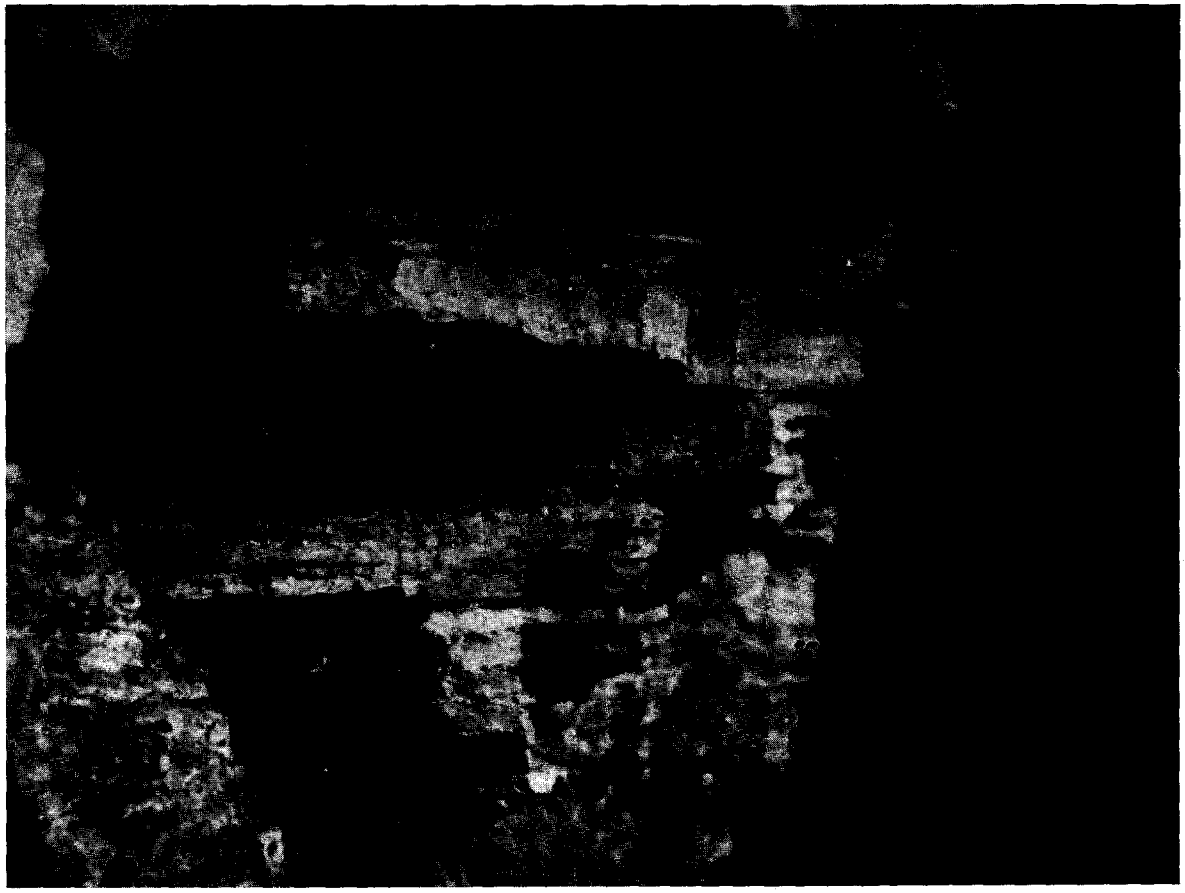
can only be interpreted on the postulate of field reversal. If the very convincing argument put forward by Bruckshaw and Robertson, and by Hospers is accepted this would imply at least two and probably several additional reversals of the main dipole field since Devonian times. It is conceivable however that the observed results are not due to reversal of the main field, but to chemical and physical changes which have occurred in the rocks since compaction.

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Exposure of sandstones of the Keuper Marl Series on the bank of the river at Radcliffe-on-Trent, near Nottingham (site 8).